Test report 1: Kernel, language, engine, CLI

controller-cad · As of 2026-10-23 01:24 · Commit fd8dda6 2026-10-23 01:24 Facade: keep SMLib's own result when the check fails (branch `worktree-agent-a45db6d1ee5be50fc`)

202	33/33	1.58 MiB	141 ms
Rust tests passed (0 failed, 0 ignored)	OpenSCAD models within tolerance	Wasm total, Brotli (SMLib 1.24 MiB, engine 0.34 MiB)	Source → root mesh (example gehaeuse, Node, cold)

1 Scope

This report covers the four lower layers of the application (the UI is not part of this report):

- **Kernel:** SMLib/SM_API (NVIDIA usd-brep @ c9c5979) behind a C++ facade with a flat C ABI (handles, error codes, face tags for provenance), linked natively and as Emscripten Wasm (`smlib.wasm`); Rust binding `kernel-smlib`. Operations: primitives, transformations, booleans, fillet/chamfer, extrusion (with twist/scale), revolve, loft, sweep, NURBS surface, thicken, shell, edge/face queries, classification, tessellation, mass properties.
- **Language:** OpenSCAD dialect (`cad-syntax`, `cad-eval`): variables, functions, modules, `children()`, `for` / `if` / `let` / `each`, list comprehensions, `echo` / `assert`, modifiers, 2D/3D primitives, extrusions; plus the extensions from the spec: grids (`grid`), anchors, named instances, `box()` between grid points, fillets/chamfers with zones, loft, sweep, handles.
- **Engine:** `cad-engine` evaluates the geometry tree incrementally with a cache (content-based keys), maps faces and edges to source lines, selects edges via zones and exports STL/3MF; `cad-session` provides the protocol for the worker and the CLI.
- **CLI:** `cad check | eval | inspect | render | export` with `--json`; `render` uses the software rasterizer `cad-render`.

2 Test environment

System	Ubuntu 24.04.5 LTS (Linux 6.18.44-fc-v140), Intel(R) Xeon(R) Processor @ 2.80GHz, 4 cores, 16 GiB RAM, container without GPU
Rust	rustc 1.97.0 (2d8144b78 2026-07-07); cargo 1.97.0 (c980f4866 2026-06-30)
Kernel	usd-brep @ c9c5979 (SMLib, SM_API), prebuilt natively: /home/user/controller-cad/smlib/build-native
Emscripten	emcc (Emscripten gcc/clang-like replacement + linker emulating GNU ld) 6.0.13 (e13ee988ce3d87a738dd8f3661d02223308c412a)
wasm-bindgen	wasm-bindgen 0.2.129
Node	v22.22.0
OpenSCAD (reference)	OpenSCAD version 2021.01; images with xvfb-run, orthographic
PDF	Chromium from Playwright 1.56.1
Python	3.13.16

Release build of the CLI: 0.1 s (incremental). All numbers come from one run of `tools/report/report1.py`; times are wall-clock times on this machine.

3 Test results

3.1 Rust tests per crate

Crate	Content	passed	failed	ignored	Test time	cargo run time	Status
`cad-cli`	CLI	7	0	0	2.91 s	4.0 s	ok
`cad-kernel`	Kernel trait, mesh, export	0	0	0	0 s	0.2 s	ok
`cad-render`	Software rasterizer	4	0	0	1.50 s	2.4 s	ok
`cad-engine`	Evaluation, cache, provenance, zones	31	0	0	6.99 s	8.0 s	ok
`cad-ir`	Geometry tree, cache keys	0	0	0	0 s	0.2 s	ok
`cad-syntax`	Lexer, parser, AST	16	0	0	0.00 s	0.2 s	ok
`cad-eval`	Language evaluation, grids, anchors, handles	102	0	0	0.46 s	0.8 s	ok
`kernel-smlib`	SMLib facade (native)	31	0	0	6.77 s	7.3 s	ok
`cad-session`	Session, protocol	8	0	0	0.13 s	1.5 s	ok
`cad-wasm`	Wasm entry point (worker)	0	0	0	0 s	0.1 s	ok
`compat-tests`	OpenSCAD comparison, stage 1	3	0	0	0.04 s	0.2 s	ok
`kernel-mock`	Test kernel	0	0	0	0 s	0.1 s	ok
Total		**202**	**0**	**0**	**18.80 s**	**25.2 s**	

Test time: sum of the times reported by the test binaries (debug build, SMLib and facade optimized). cargo run time: `cargo test -p <crate>` after a prior build.

3.2 Repeated runs of `kernel-smlib/tests/kernel.rs`

Mode	Runs	failed	Test	Duration
serial (--test-threads=1)	30	0	–	201 s

parallel (default, 4 threads)	30	0	–	201 s
parallel (`--test-threads=8`)	30	0	–	203 s

Before the fixes: 4 of 40 runs with 8 threads failed (shell too thick). After the rand() fix, 100 runs with 8 threads still failed once each in `invalid_profiles_fail_cleanly` (memory error in SMLib's shell) and `handles_are_released` (parallel counter); both have been fixed since.

3.3 Kernel as Wasm (Node)

`node crates/kernel-smlib/js/test-node.mjs` : 10 of 10 tests passed (ok).

Test	Time	Status
box volume	23.1 ms	ok
box minus cylinder + provenance	88.1 ms	ok
fillet all edges of a box	330.8 ms	ok
fillet too large → error with edges	111.3 ms	ok
classify	1.9 ms	ok
tessellate	42.4 ms	ok
extrude, revolve, mirror	76.9 ms	ok
union of three orthogonal cylinders (checked boolean)	497.3 ms	ok
twisted extrude: rotate, then scale (OpenSCAD)	64.8 ms	ok
invalid handle	0.3 ms	ok

3.4 Bugs fixed in this round

Bug	Cause	Fix, evidence
Union of three orthogonal cylinders: 3237 instead of 3800 mm³ (05_nested_booleans)	For (Z ∪ Y) ∪ X, SMLib reports success but drops the −x arm of the X cylinder (bounding box x ≥ −4); (Z ∪ X) ∪ Y is correct, (Y ∪ X) ∪ Z ends with SM_ERR. Only with equal radii (intersection curves with touching points); with r = 3.9 or 4.1 for the third cylinder the result is correct. `mass_props` and the mesh are equally wrong: not a tessellation or export bug. Pairwise, the facade already computed correctly.	The facade checks every pairwise result (volume inequalities, location of probes, 5 per face, just inside and outside the operand faces) and otherwise tries: swap operands, different order, and finally shift the second operand by 10^{-6}…10^{-4} of the model size; if nothing passes, SMLib's result is kept as before. Tests: `kernel.rs` and `engine.rs::union_of_three_orthogonal_cylinders` (all 6 orders, nested, as a hole), Node test.
linear_extrude with twist: bounding box off by 1.25 mm (12_linear_extrude_twist)	Section t was scaled first and rotated afterwards; OpenSCAD rotates first and then scales along x/y (`Scaling(s)·Rotation2D(-twist·t)`). With `scale = [1, 0.5]` this gives different cross-sections (y up to 5.08 instead of 3.83 mm). Direction of rotation and start angle were correct. In addition: sections every 22.5° (bounding box error $3{\cdot}10^{-3}$ mm), and the mesh was 2.9 % above the volume because SMLib limits the chord height only at triangle edges.	Order as in OpenSCAD; sections every 7.5° (bounding box within 10^{-4} mm, volume 540.000 of 540). On freeform surfaces the tessellation measures the deviation of the triangle centers and refines until ≤ 2·tol (at most 3 passes, triangle budget). Tests: `twisted_extrude_with_nonuniform_scale_matches_openscad`, `twisted_extrude_mesh_volume`, `engine.rs::linear_extrude_twist_nonuniform_scale_like_openscad`.
Flaky test `invalid_profiles_fail_cleanly`	Two causes, both in the too-thick `shell` (10 mm cube, 6 mm wall, must fail). (1) SMLib calls `srand(time(NULL))` in `SmFace::GetPointsInFace` and `SmShell::IsInnerShellOf`: before, 4 of 40 runs (8 threads) failed; with a fixed clock (`time()` via LD_PRELOAD), 0 of 40. An `srand` before each operation is not enough, because SMLib reseeds in the middle of the operation. (2) After that, still 1 of 100: with a too-thick wall, SMLib's shell reads freed memory (valgrind: *Invalid read* in `SmProtoEdge::RemoveProtoVertex`; solidcore UB-239/242); depending on the heap it fails, returns a body, or crashes (segfault in a loop after 9 calls).	(1) The facade defines `rand()` / `srand()` itself (statically linked, native and Emscripten): fixed seed, reset on every `sc_*` call. (2) `sc_shell` rejects walls that do not fit between two opposite planar faces (distance ≤ 2·t) before SMLib is called. In addition, `handles_are_released` failed when run in parallel (process-wide counter): separate test binary. Section 3.2; tests `determinism.rs`, `handles.rs`.
Diagonal bands on subtracted bodies rendered transparent	`cad-render` sorted transparent triangles by centroid. Where triangles overlap in depth, especially on coplanar faces (subtracted body and its cavity in the dimmed result), the order flips along the triangle diagonals.	Fragments are sorted by depth per sample, coplanar ones in scene order (figure *gehaeuse --line 12* in section 5). Test: `render.rs::transparent_coplanar_faces_have_no_diagonal_bands`.

4 Comparison with OpenSCAD

`tests/openscad-compat/compare.py` exports each model as STL (`cad export`, chord tolerance 0.01 mm) and compares volume, area and bounding box with OpenSCAD 2021.01 (`$fn = 128` for round shapes, see `tests/openscad-compat/README.md`). Tolerances: bounding box 0.01 mm; volume and area 1 % (*polygonal*) or 2 % (*rounded*). Result: **33 of 33** within tolerance, run time 11.3 s.

Model	Class	V OpenSCAD [mm³]	V ours [mm³]	ΔV	ΔA	ΔBBox [mm]	Triangles	Time	Status
`01_cube_variants`	polygonal	3 208.0	3 208.0	0.00 %	0.00 %	0.0000	60	0.05 s	ok
`02_sphere_variants`	rounded	1 016.9	1 013.7	0.31 %	0.15 %	0.0015	14 912	0.81 s	ok
`03_cylinder_variants`	rounded	2 104.0	2 101.5	0.12 %	0.05 %	0.0000	1 260	0.18 s	ok
`04_cone_variants`	rounded	1 342.0	1 340.8	0.09 %	0.05 %	0.0000	1 010	0.12 s	ok
`05_nested_booleans`	rounded	8 310.3	8 312.7	0.03 %	0.02 %	0.0000	6 688	1.52 s	ok
`06_hole_grid`	rounded	6 295.6	6 296.7	0.02 %	0.01 %	0.0000	6 252	1.46 s	ok
`07_crown_ring`	rounded	2 160.7	2 160.9	0.01 %	0.02 %	0.0000	1 088	0.83 s	ok
`08_module_children`	rounded	6 172.5	6 171.6	0.02 %	0.00 %	0.0000	2 072	0.44 s	ok
`09_rotate_mirror`	rounded	1 460.1	1 459.9	0.01 %	0.01 %	0.0000	324	0.07 s	ok
`10_scale_multmatrix`	polygonal	3 400.0	3 400.0	0.00 %	0.00 %	0.0000	48	0.05 s	ok
`11_linear_extrude_basic`	polygonal	1 513.0	1 513.0	0.00 %	0.00 %	0.0000	32	0.02 s	ok
`12_linear_extrude_twist`	rounded	2 688.9	2 660.9	1.04 %	1.49 %	0.0000	15 730	0.54 s	ok
`13_rotate_extrude_ring`	rounded	1 307.1	1 306.3	0.06 %	0.01 %	0.0000	17 120	0.66 s	ok
`14_pipe_bend`	rounded	1 573.9	1 572.4	0.09 %	0.03 %	0.0000	9 672	0.64 s	ok
`15_vase`	rounded	6 362.3	6 362.3	0.00 %	0.00 %	0.0000	2 556	0.20 s	ok
`16_polygon_holes`	polygonal	2 318.0	2 318.0	0.00 %	0.00 %	0.0000	80	0.04 s	ok
`17_polyhedron`	polygonal	626.7	626.7	0.00 %	0.00 %	0.0000	14	0.01 s	ok
`18_functions_let`	polygonal	7 040.0	7 040.0	0.00 %	0.00 %	0.0000	52	0.05 s	ok
`19_hex_nut`	rounded	264.7	264.9	0.07 %	0.01 %	0.0000	552	0.11 s	ok
`20_box_with_lid`	rounded	20 574.8	20 576.1	0.01 %	0.01 %	0.0000	2 284	0.76 s	ok
`21_angle_bracket`	rounded	10 408.0	10 408.3	0.00 %	0.00 %	0.0000	1 088	0.32 s	ok
`22_gear`	rounded	6 943.8	6 944.3	0.01 %	0.00 %	0.0000	1 616	0.37 s	ok
`23_shelf_holder`	rounded	10 423.8	10 424.1	0.00 %	0.00 %	0.0000	1 184	0.23 s	ok
`24_modifiers`	rounded	3 622.5	3 623.2	0.02 %	0.00 %	0.0000	284	0.06 s	ok
`25_root_modifier`	rounded	346.5	346.4	0.05 %	0.00 %	0.0000	268	0.06 s	ok
`26_if_else_variants`	rounded	1 661.6	1 660.5	0.07 %	0.04 %	0.0012	4 380	0.34 s	ok
`27_spacers_2d`	rounded	2 961.5	2 961.4	0.00 %	0.01 %	0.0000	2 512	0.47 s	ok
`28_cable_clip`	rounded	560.9	561.0	0.02 %	0.01 %	0.0013	796	0.20 s	ok
`29_assert_strings`	rounded	1 907.4	1 907.5	0.00 %	0.00 %	0.0000	276	0.08 s	ok
`30_each_ranges`	polygonal	832.0	832.0	0.00 %	0.00 %	0.0000	116	0.07 s	ok
`31_phone_stand`	polygonal	40 840.2	40 840.2	0.00 %	0.00 %	0.0000	84	0.03 s	ok
`32_low_fn_shapes`	polygonal	1 199.0	1 199.0	0.00 %	0.00 %	0.0000	102	0.04 s	ok
`33_global_fn`	rounded	1 625.6	1 627.4	0.11 %	0.04 %	0.0060	4 860	0.35 s	ok

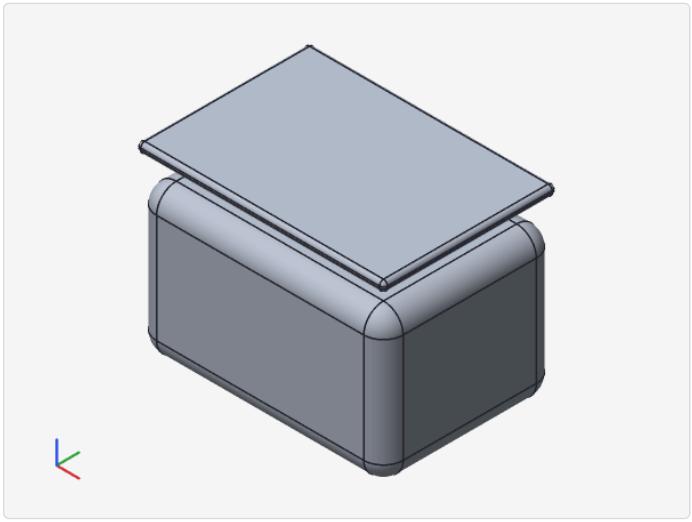
> **On 12_linear_extrude_twist:** here OpenSCAD's own mesh deviates from the exact body. The exact value is 1280 + 840 + 540 = 2660 mm³ (twisted cross-sections keep their area); OpenSCAD gives 2 688.9 mm³ because it splits every twisted side face into two planar triangles with a fixed diagonal. Our STL has 2 660.9 mm³. The 1.04 % deviation in the table is therefore OpenSCAD's faceting.

5 Renderings

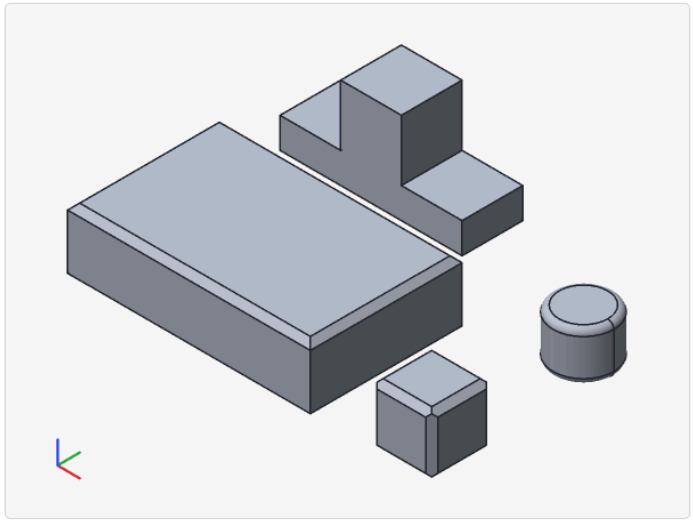
All images 600×450 px, isometric view (looking along (−1, 1, −1), orthographic), `cad render <file> --size 600x450`. OpenSCAD images: `openscad -o x.png --render --imgsize=600,450 --viewall --autocenter --projection=ortho --camera=100,-100,100,0,0,0` with the same `-D` values as the reference.

5.1 Examples from the spec (`examples/spec/`)

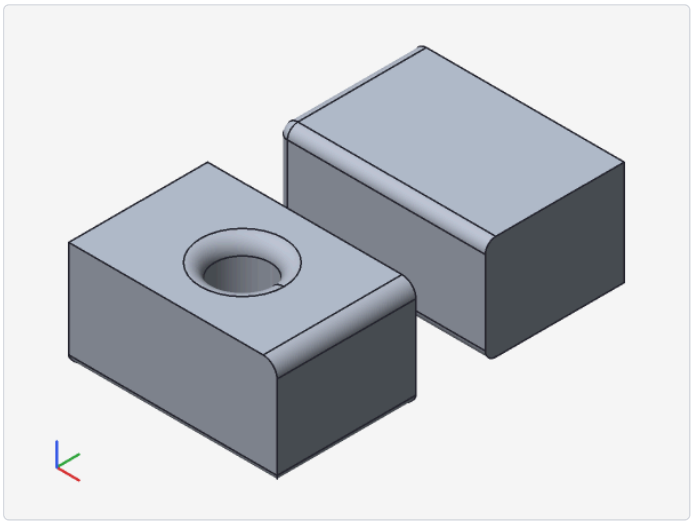
Only with our CLI: grids, anchors, fillets and zones do not exist in OpenSCAD.



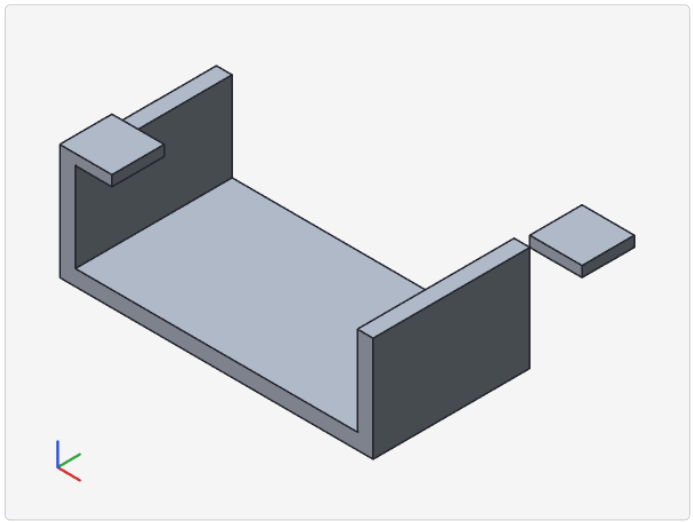
dose_shell – shell(t) { body; zone } – can made from a filleted solid ($fillet), opened at the top via the zone at the grid plane deckel; anchor for the lid.



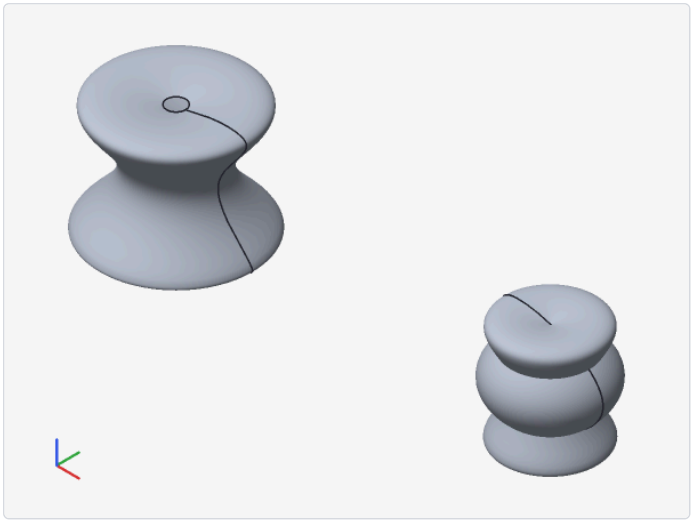
fase_chamfer – chamfers in all three forms – chamfer block with zone, union(chamfer = ...) and $chamfer on primitives; plus $fillet for a rounded knob.



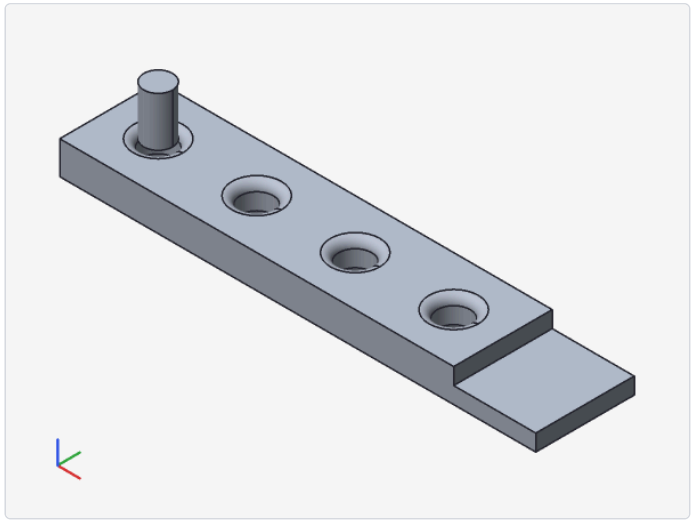
fillet_zonen – zone block fillet(r) { body; zone; ... } with grid line, grid plane and volume zone, select = "touch", nested blocks with two radii.



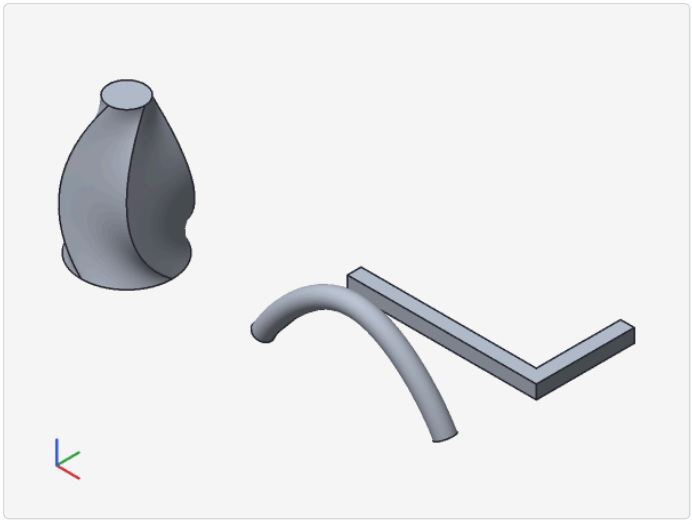
gehaeuse – spec example "gehaeuse" (enclosure) – grid g in the module, box between grid points, anchors facing outward, named instance k, lid at the anchor and a second grid from k.ecke.x.



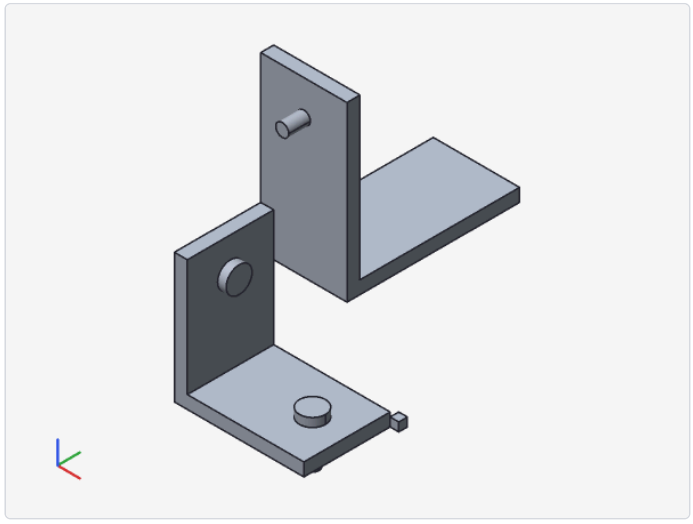
knauf_mirror – knob – 2D grid for a spline profile, rotate_extrude, torus, difference(fillet = ...) at the hole; mirrored copy at a plane of a scaled grid.



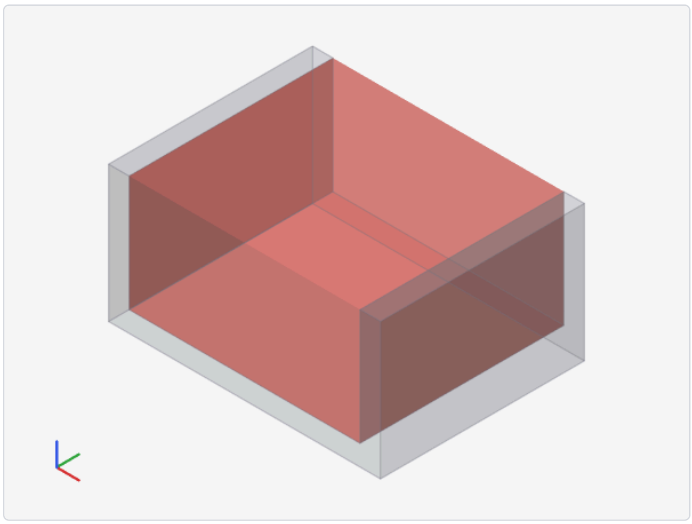
lochplatte – perforated plate with indexed grid planes (lists), holes at grid points, a second grid placed at grid points and difference(fillet = 1) for the hole edges.



loft_sweep – profile operations – loft from circle to square (funnel), sweep of a circle along a spline path (handle bar) and along a linear path made of grid points.



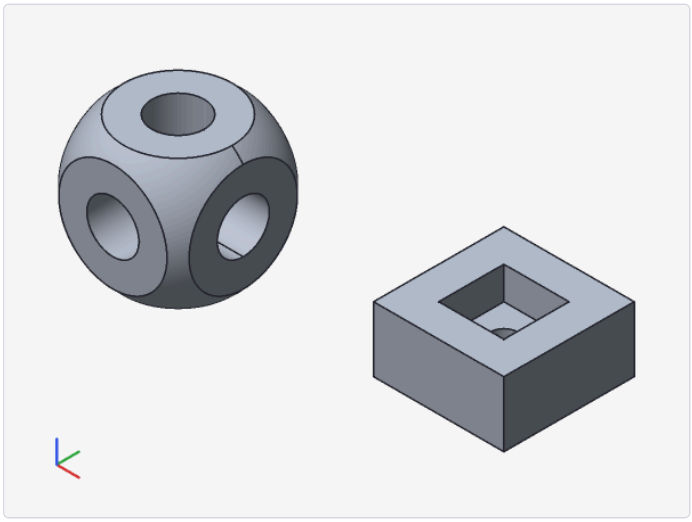
winkel_anker – anchors from outside – two named instances of an angle bracket, screws at the anchors with rotate(to = dir(k.a)), rotated instance, grid from anchor coordinates.



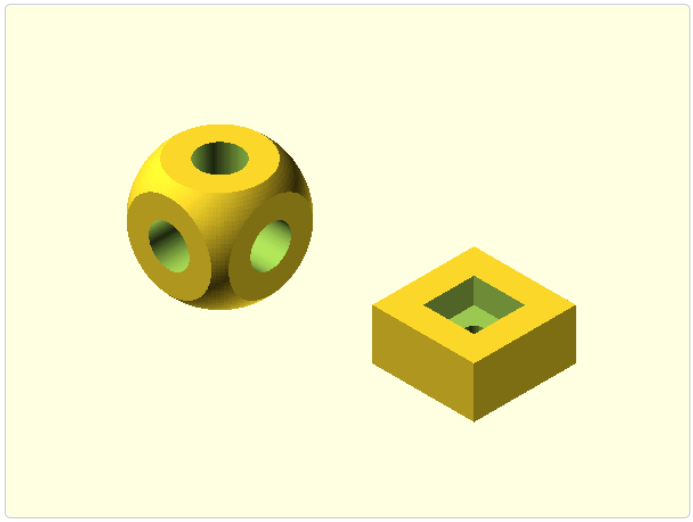
gehaeuse --line 12 – line 12: subtracted body red and transparent, the rest dimmed

5.2 OpenSCAD models: ours (left) and OpenSCAD (right)

`05_nested_booleans` nested booleans – cube intersected with a sphere, cross hole subtracted.

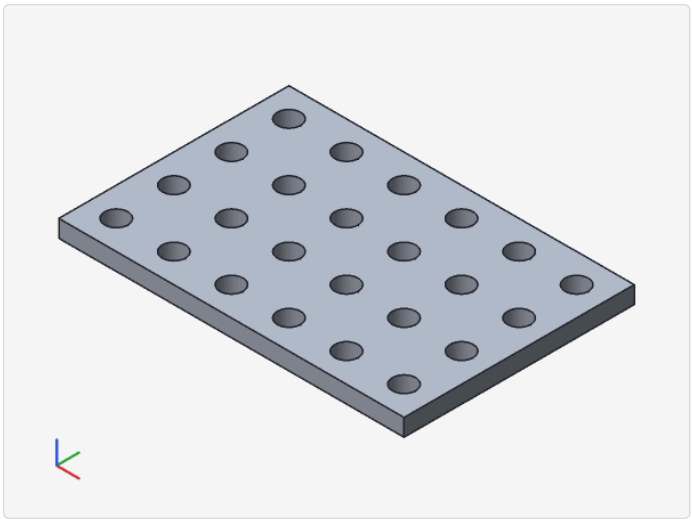


cad render

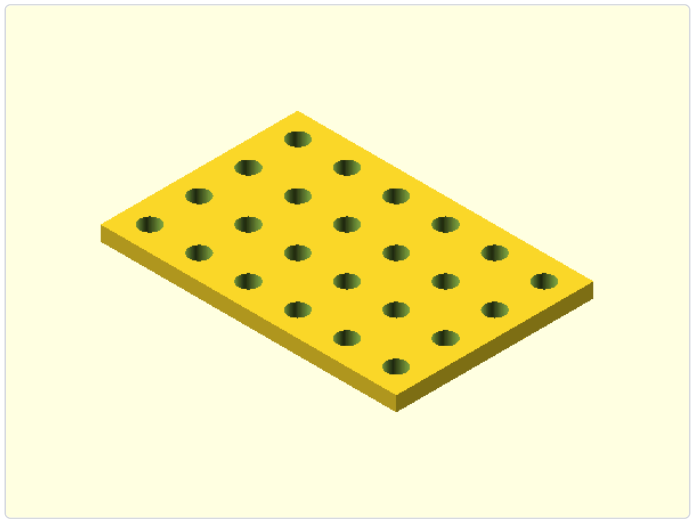


OpenSCAD 2021.01

`06_hole_grid` perforated plate – a for loop with two variables creates a grid of holes.

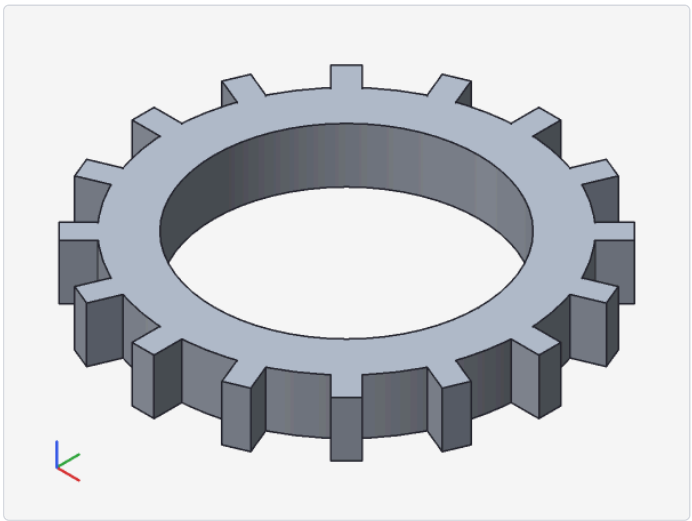


cad render

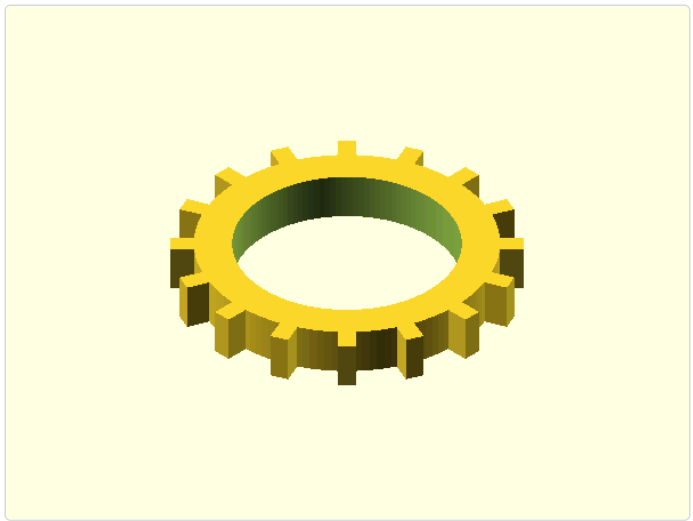


OpenSCAD 2021.01

`07_crown_ring` crown-gear-like ring – teeth evenly distributed with for and rotate.

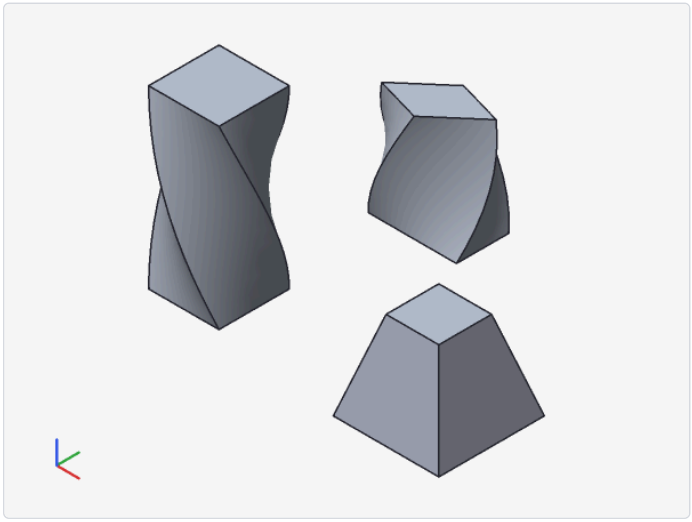


cad render

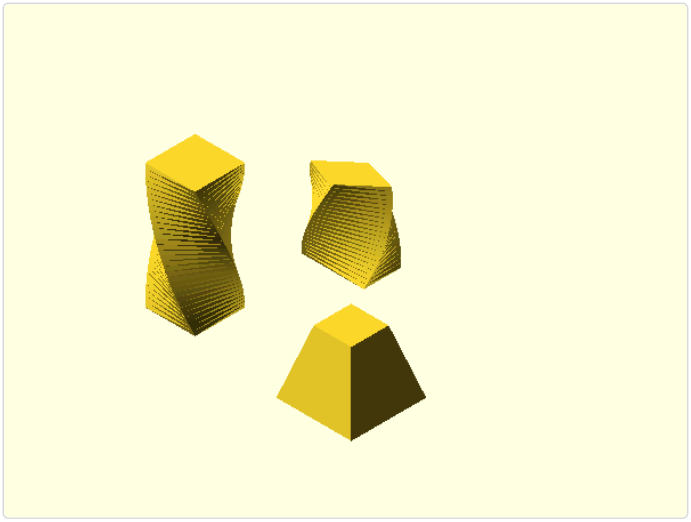


OpenSCAD 2021.01

`12_linear_extrude_twist` linear_extrude with twist and scale – twisted bar and tapered base.

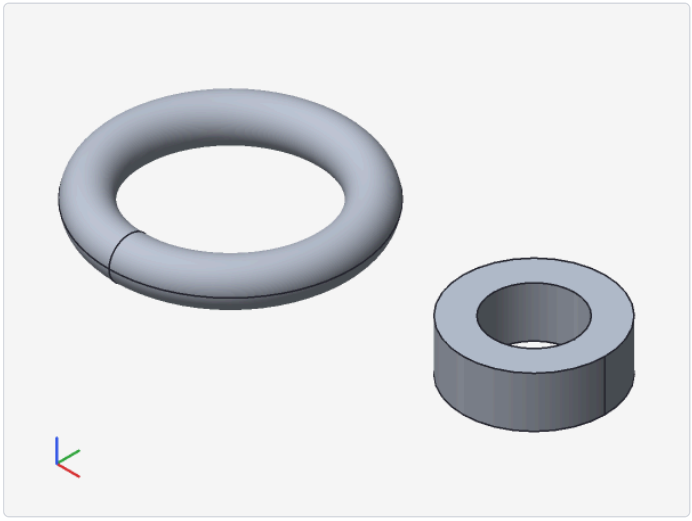


cad render

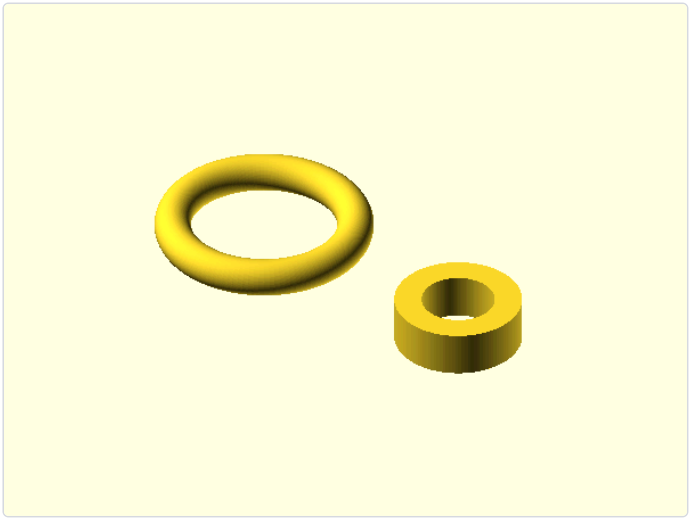


OpenSCAD 2021.01

`13_rotate_extrude_ring` rotate_extrude of an offset circle (ring) and of a rectangle (disc with hole).



cad render

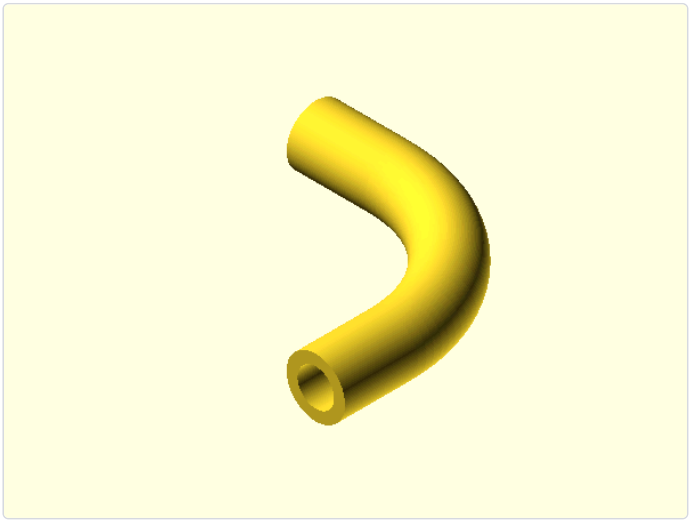


OpenSCAD 2021.01

`14_pipe_bend` pipe bend via rotate_extrude(angle = 90) with two straight connecting pieces.

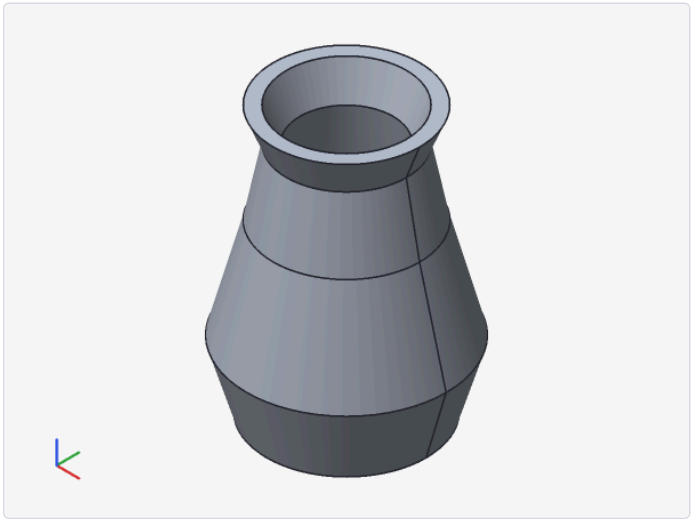


cad render

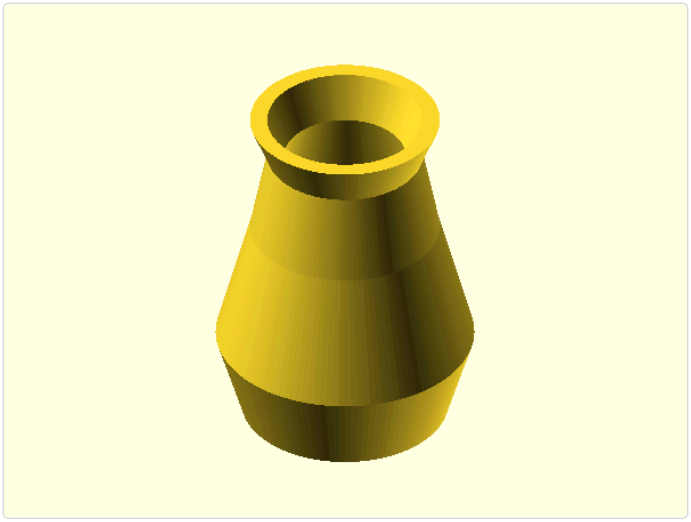


OpenSCAD 2021.01

`15_vase` **vase as rotate_extrude of a polygon profile (wall 2 mm, bottom 2 mm).**

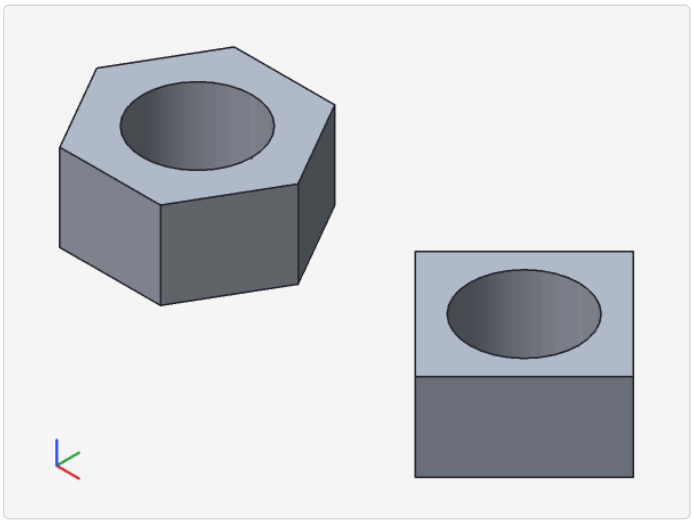


cad render

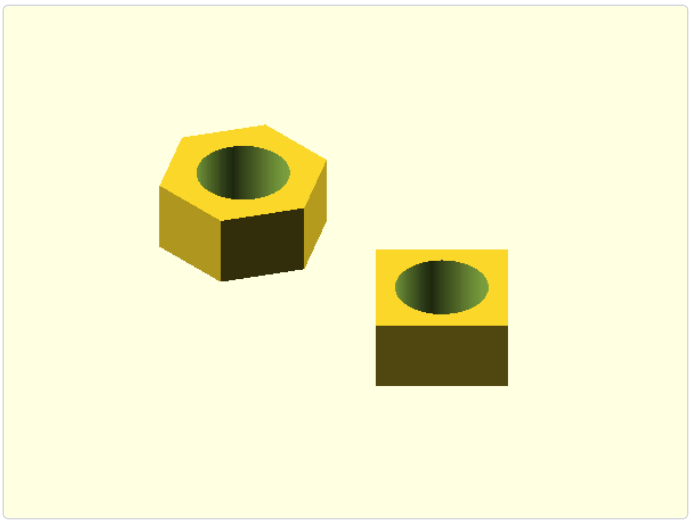


OpenSCAD 2021.01

`19_hex_nut` **M5 hex nut – $fn = 6 at the call creates the hexagon, the hole is round.**

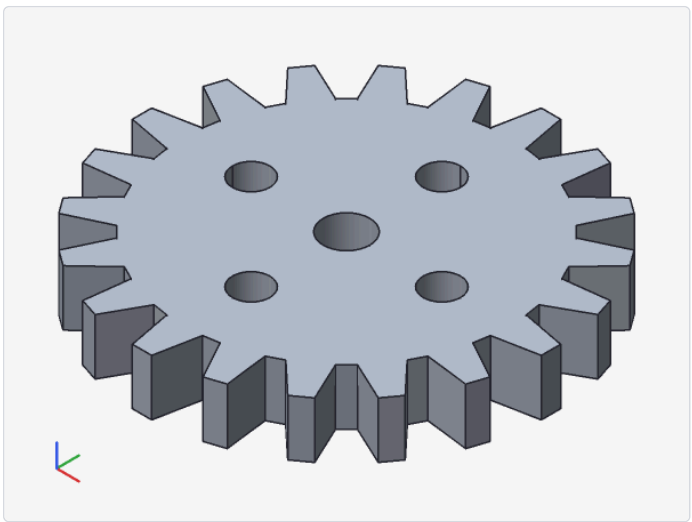


cad render

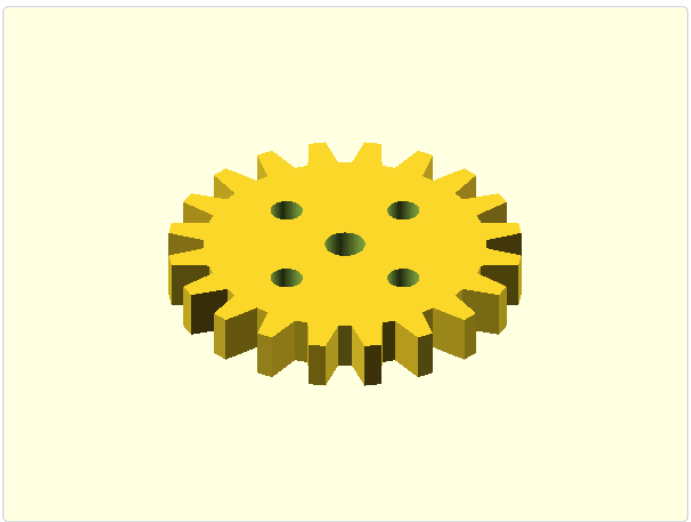


OpenSCAD 2021.01

`22_gear` **simplified spur gear – trapezoidal teeth as a polygon from a list comprehension.**

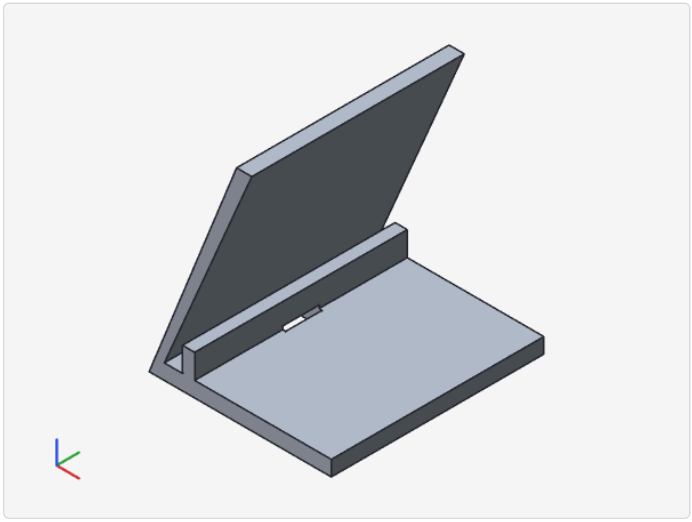


cad render

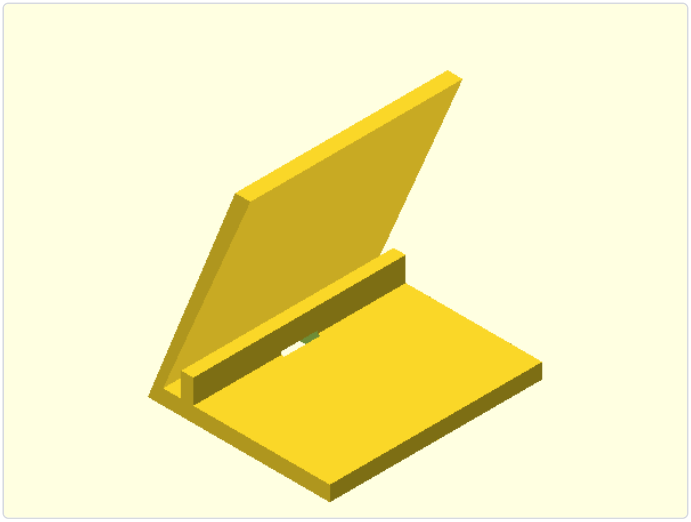


OpenSCAD 2021.01

`31_phone_stand` phone stand – inclined back rest via rotate, lip, cable slot; profile via polygon.



cad render



OpenSCAD 2021.01

6 Measurements

6.1 Wasm sizes

File	raw	gzip −9	Brotli q11
`smlib.wasm`	4 993 KiB	1 769 KiB	1 270 KiB
`smlib.js`	14 KiB	5 KiB	4 KiB
`cad_wasm_bg.wasm`	1 186 KiB	464 KiB	353 KiB
`cad_wasm.js`	28 KiB	6 KiB	5 KiB

`smlib.wasm`: SMLib + SM_API + facade, Emscripten `-O3 -fwasm-exceptions`; `cad_wasm_bg.wasm`: language, engine, session (wasm-bindgen, wasm-opt −O3).

6.2 Load time and evaluation (Node 22, one process as in the worker)

`web/scripts/smoke-node.mjs --json` with the example `gehaeuse`, median of 3 runs, step budget 16 ms.

Step	Time
Load, compile, initialize smlib.wasm	32.4 ms
Load and initialize cad_wasm	32.4 ms
SetSource → preliminary scene (without kernel)	20.7 ms
SetSource → complete scene	123.1 ms
SetSource → root mesh	141.3 ms
Request meshes of all nodes	6.1 ms
Drag handle → TextEdit → new scene	12 ms
… → coarse root mesh	14.1 ms
Repeated SetSource (warm cache) → preliminary scene	0.9 ms
Repeated SetSource (warm cache) → scene	21.9 ms
Repeated SetSource (warm cache) → root mesh	23.9 ms
Evaluation steps until the complete scene	5
Triangles of root mesh / nodes of the scene	40 / 8

Kernel only (`test-node.mjs`): read file 4.5 ms, compile, instantiate and initialize 29.0 ms. Scenario: box-cylinder + fillet(5 top edges) + tessellate (1376 triangles): primitives 4.1 ms, boolean 24.4 ms, edges 0.5 ms, fillet 64.3 ms, tessellate 44.3 ms; total incl. load 1410 ms.

6.3 Native evaluation (CLI)

`cad export` per model (process start, evaluation, tessellation, STL): median 0.20 s, total 11.2 s; longest `05_nested_booleans` 1.52 s, `06_hole_grid` 1.46 s, `07_crown_ring` 0.83 s.

`cad render` of the spec examples (600×450, 2×2 supersampling): dose_shell 0.69 s, fase_chamfer 0.26 s, fillet_zonen 0.31 s, gehaeuse 0.06 s, knauf_mirror 2.67 s, lochplatte 0.51 s, loft_sweep 0.47 s, winkel_anker 0.46 s.

7 Known limitations

- SMLib booleans are unreliable in special configurations. The facade detects wrong results by sampling, not completely: an error between all probe points goes undetected. The check uses SMLib's point classification, which itself is wrong for rotated spline bodies (knauf_mirror); if no variant passes, SMLib's own result is therefore kept. In the shift fallback, the second operand deviates by up to 10^{-4} of the model size. The check costs time: `cad`

`export` of all 41 examples, measured during the fix, 13.7 s → 18.9 s (+38 %), most for unions of many parts (07_crown_ring ×2.1) and for knauf_mirror (×1.9, fallbacks).

- Open results: if bodies touch along an edge, the union in SMLib is not a closed body; `cad` then reports volume 0 (winkel_anker; the STL is fine). knauf_mirror: the rotated spline body minus the hole is open, torus and pin are missing from the result, the fillet at the hole fails (SM_ERR) – all as before this round, still open.
- Tessellation of freeform surfaces: the refinement sets a global maximum edge length, so planar faces of the same body also become finer (12_linear_extrude_twist: 15 730 triangles, OpenSCAD 452).
- Twist extrusion is a cubic loft approximation through sections every 7.5°. Circular profiles with non-uniform `scale` and `scale = 0` (apex) are not supported.
- OpenSCAD facets, we compute exactly: volumes of round parts deviate systematically by 0.01–0.3 %; with twist, OpenSCAD itself is off by ~1 % (section 4).
- Determinism: `rand()` is replaced with a fixed, process-wide implementation. Memory errors in SMLib make results depend on the heap: the shell pre-check only covers planar, opposite walls; walls that barely fit (10 mm cube, t = 4.99) already show invalid reads under valgrind. Only the upstream fixes (tangency-labs/smlib 93efd6d, af8db37) or solidcore will resolve this.
- `cad render` is a software rasterizer (orthographic, Lambert, 2×2 supersampling) without shadows or perspective. Per-pixel transparency needs memory proportional to the number of layers.
- Not part of this report: browser UI, STEP export, switch to solidcore (M8).

Generated with `python3 tools/report/report1.py` (PDF: `tools/report/print-pdf.mjs`, Chromium from Playwright).